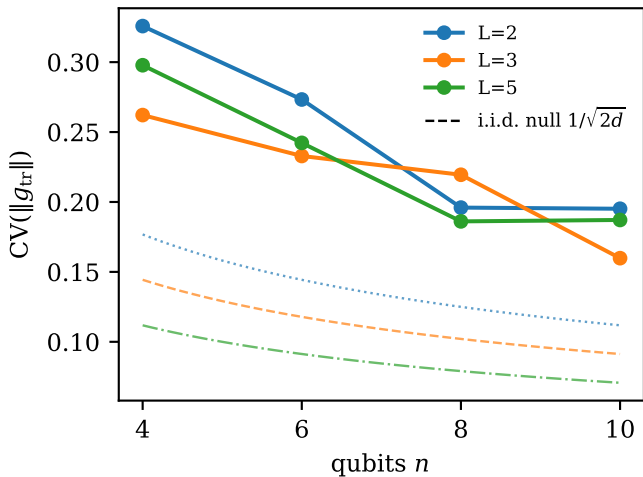
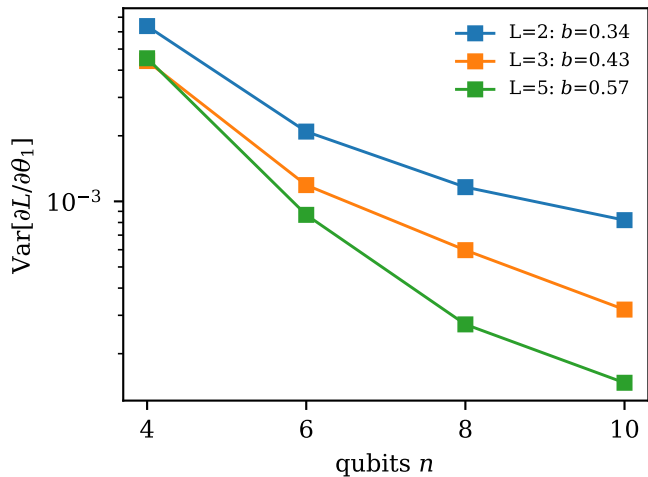
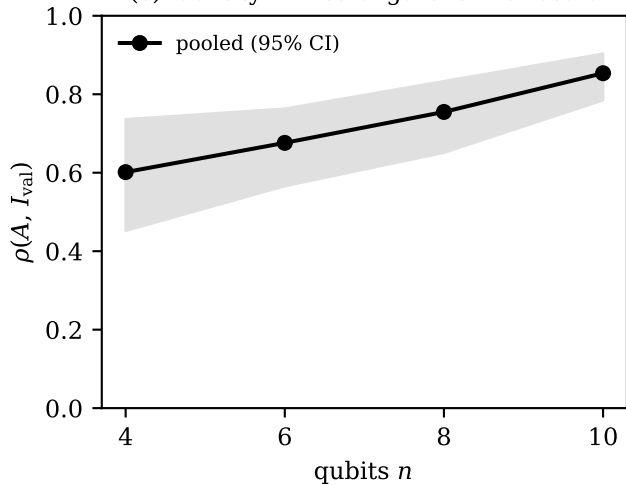


(a) gradient-norm concentration

(b) barren-plateau probe: $\propto e^{-bn}$ 

(c) identity link strengthens with scale



(d) gap association vanishes under joint control

